

Evolutionary Methods in Collective Herding Systems

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1 INTRODUCTION

Multi-Agent Systems (MAS) are composed of agents that decide on a path of action based on the information available to them regarding the environment and other agents in a system [4]. These systems are an important branch of Artificial Intelligence and are well suited for applications that require parallel or decentralized computation or inter-component communication [4]. We will be addressing two significant areas of research in MAS: body-brain evolution in swarm robotics, and hybrid evolutionary methods in MAS.

Environmental complexity appears to play a significant role in the benefits of body-brain co-evolution. Studies show that complex environments tend to select for greater behavioural and morphological diversity [2]. This aligns with biological observations where environmental heterogeneity promotes phenotypic diversity [22]. This project will use a simulated herding task as a case study to explore the impact of environmental complexity on body-brain evolution in a multi-agent system. The research aims to compare the performance of robots with evolving morphologies to those with fixed morphologies in a collective herding task, and to evaluate the resulting morphologies and behaviours evolved in environments of varying complexities.

Research has shown that in MAS, hybrid evolutionary algorithms that balance both exploration of different solutions and exploitation of successful ones result in optimal system performance, compared to traditional objective-based algorithms or purely explorative algorithms [11]. Such hybrid algorithms have been tested on numerous different static applications, such as two-dimensional maze navigation tasks and locomotion tasks [6, 8, 21]. However, there is relatively little research that compares the effectiveness of hybrid evolutionary methods in achieving an optimal balance between exploration and exploitation within dynamic environments. This project will contribute to this area of research by evaluating the effectiveness of two hybrid evolutionary algorithms, Novelty-Surprise Search with Local Competition (NSS-LC) and Dominated Novelty Search (DNS), against an objective-based evolutionary algorithm, Steady-State Genetic Algorithm (SSGA), on a sheep-herding task in three dynamic maze environments.

2 PROBLEM STATEMENT

A key aspect of MAS is the way in which agents evolve and find solutions and which solutions are selected. In particular, the balance between exploring different possible solutions and exploiting successful solutions is crucial to the success of the system.

Over the years, numerous different methods have been used to evolve agents in MAS, ranging from purely objective-based methods, which use an objective function that rewards getting closer to the goal, to purely explorative methods, such as Novelty Search and ultimately hybrid methods, that do both exploration and exploitation, such as Novelty Search with Local Competition (NSLC) and

Multi-dimensional Archive of Phenotypic Elites (MAP-Elites). The evolution away from purely objective-based algorithms, such as SSGA, is due to the limitations these algorithms face, such as being deceived by local optima [4]. Hybrid exploration-exploitation methods are able to solve much more complex and deceptive problems than objective-based methods or purely explorative methods, due to their ability to avoid getting stuck in local optima and finding the globally optimal solution. While hybrid evolutionary algorithms have been widely tested in MAS, most research has focused on static environments, such as testing these algorithms using simulated wheeled robots to navigate mazes of varying complexity [6, 21]. Consequently, the effectiveness of hybrid evolutionary methods in dynamic environments remains an underexplored area of research.

The evolution of both morphology and control is an essential aspect of autonomous and adaptive robotic system design. Embodied artificial intelligence considers the evolutionary system as a whole and asserts that intelligence emerges from interactions between an agent's morphology and its environment, contrasting with traditional AI approaches that treat intelligence as purely computational [20]. Prior studies suggest that more complex environments tend to select for greater behavioural and morphological diversity [2] and that co-evolution can be beneficial when solving complex tasks [18]. Yet, the role of the environment in the evolutionary process remains insufficiently studied.

2.1 Research Questions

The research questions addressed by this project are summarised below.

- (1) How does the performance of agents subjected to simultaneous body-brain evolution compare to that of agents with fixed morphology?
- (2) What is the impact of environmental complexity on body-brain evolution in collective herding tasks?

The agent implementations used to answer research questions 1 and 2 are detailed in Section 3.1, while Section 3.2 describes the morphological evolution of the dog agent. Furthermore, the controllers will be evolved using SSGA, which is described in Section 3.3.1.

- (3) Which of the algorithms (NSS-LC, DNS and SSGA) achieves the ideal balance between exploration (maze coverage) and exploitation (sheep captured) in dynamic collective herding tasks?
- (4) How do the hybrid algorithms NSS-LC and DNS compare to each other and to the exploitation-only SSGA, in terms of exploration, exploitation, and overall effectiveness in dynamic collective herding tasks?

The sheep and dog agents used in the collective herding task to address research questions 3 and 4 are described in Sections 3.1.1 and 3.1.3, and the algorithms that will be used (NSS-LC, DNS and SSGA) are explained in Section 3.3.

An overview of the experiments to be conducted is provided in Section 3.5 and the experimental specifics are outlined in Table 1. The allocation of work related to answering these research questions is detailed in Section 7.6.

3 METHODS AND EXPERIMENTS

This section outlines the experimental setup used to evaluate body-brain evolution and hybrid evolutionary algorithms in a collective herding task. It describes the agents involved, the evolutionary methods for dog morphology and behaviour evolution, the task environments, the proposed experiments, and the evaluation metrics.

3.1 Agents

3.1.1 Dog Agents - Fixed Morphology. These robot agents are subjected to neuro-evolution. They utilise a simple, circular morphology as in e-puck robots [9]. The dog controllers are implemented by a fully connected feed-forward Artificial Neural Network (ANN) adapted with the relevant evolutionary algorithms for the collective herding task [10]. The ANN uses *tanh* activation functions between network layers and has a simple architecture with 9 sensory input nodes, 10 hidden nodes and 2 motor output nodes which correspond to 110 connection weights for the genome [9]. The 9 input nodes comprise distance and angle readings from 3 radar sensors (one per object type) and the target zone. Distance and angle values are normalised in the range $[0.0, 1.0]$ and $[-1.0, 1.0]$ respectively, where 0 represents undetected and 1.0 represents closest possible distance, and -1.0 denotes -180 degrees and 1.0 denotes +180 degrees [9]. The motor outputs consist of the dog’s translation and rotation values. Translation values are normalised in the range $[-1.0, 1.0]$, with -1.0 and +1.0 indicating maximum backwards translation speed and maximum forwards translation speed respectively [10]. Rotation values are normalised in the range $[-1.0, 1.0]$, where -1.0 represents the maximum rotation speed to the left and +1.0 represents the maximum rotation speed to the right [10].

3.1.2 Dog Agents - Evolving Morphology. These robot agents undergo body-brain evolution for the herding task. These dogs utilise the same controller implementation as those with fixed morphology. For the evolution of the robot body we must extract those elements which affect the behaviour, from the structure of the robot [12]. In the context of our research we have identified these determining elements to be the *body size*, *maximum speed*, *sensor range* and *field of view (FOV)*. The original simulations define fixed values for *maximum translational speed*, *sensor range*, *left FOV* and *right FOV* [9]. We will allow these values to vary and introduce the body size parameter. Maximum translational speed will be normalised to $[0.0, 1.0]$, with 0.0 being the slowest and 1.0 being the fastest. Sensor range will be normalised to $[0.0, 1.0]$, where 0.0 and 1.0 correspond to 0px and 100px respectively. Both left FOV and right FOV will be normalised to $[0.0, 1.0]$. For left FOV 0.0 denotes 0.0 degrees and 1.0 denotes -180 degrees. Similarly, for right FOV, 0.0 denotes 0 degrees and 1.0 denotes +180 degrees. Roborobo! utilises images to directly set up robot specifications [5]. We will use BMP files with different pixel dimensions to represent the body sizes. Each of the five body sizes will be associated with an integer in the range $[-2, 2]$, where -2 and 2 denote the smallest and largest body sizes, respectively. The morphological parameters will be represented by the

last 5 values of the dog genome and evolved separately from the neural weights. This is further described in Section 3.2.

3.1.3 Sheep Agents. Sheep agents use a Boids controller, which directs their movement, causing them to roam as a herd [10]. Sheep possess the same configuration for sensory inputs and motor outputs as dogs, but differ in their field of view, which is $[-180, 180]$ and their sensor range, being 50px [9]. The Boids controller uses the rules of avoidance, coherence and alignment, where avoidance rules use proximity thresholds, and are prioritized such that the sheep avoid the dogs and then the target zone [10]. Coherence determines the speed of the sheep in steering towards one another [10]. Alignment controls the extent to which the sheep follow the average direction of the surrounding sheep [10].

3.2 Dog Morphology Evolution Method

Each individual is represented by a genome structure which will contain the neural weights and the parameters described in Section 3.1.2. These morphological parameters will be extracted to form an array of real numbers. Based on a study by Lee et.al, the array will undergo two-point crossover and one-point mutation operations to generate new arrays representing new morphologies. Given two parent genomes and a crossover point for each parent, the crossover operation will randomly execute the exchange and average functions. The exchange function exchanges the values between the crossover points, while the average function takes the average of those values. The mutation operation will add a random value $M \in [-0.1, 0.1]$ to a randomly selected parameter, except where that parameter is body size. Mutation with respect to body size will substitute the existing body size value with a randomly generated number that satisfies the body size constraints [12]. To account for the limited energy supply of real-world implementations of robots, we will introduce trade-offs between morphological parameters.

3.3 Dog Behaviour Evolution Methods

Addressing research questions 3 and 4, SSGA, NSS-LC and DNS will each be used in separate experiments to adapt the Artificial Neural Network (ANN) of the dog controllers.

3.3.1 Steady-State Genetic Algorithm. SSGA is a traditional objective-based algorithm that maintains a constant population size by removing the least fit individual every time an offspring is added to the population, thus preserving the fitter individuals and exerting high selective pressure [23]. Although this enables SSGA to outperform standard genetic algorithms in optimisation, it also leads to premature convergence (getting stuck in local optima) and a lack of diversity, due to limited exploration of the solution space [23]. For SSGA, every individual has a fitness score and fitness is assigned according to rank [23]. In this experiment, the objective of the dog agents will be to herd a flock of sheep into a target zone, with the aim of maximizing the number of sheep in the target zone.

3.3.2 Novelty-Surprise Search with Local Competition. Novelty-Surprise Search with Local Competition (NSS-LC) is a robust and efficient hybrid evolutionary algorithm that uses NSGA-II to search for nondominated solutions based on local competition and a weighted

combination of novelty and surprise [8]. Novelty measures deviation from previously observed behaviours, while surprise measures deviation from predicted future behaviours, introducing a time-based dimension and awareness of behavioural trends [8]. Surprise predicts future behaviours based on recent generations, favouring individuals whose behaviours break these expectations [8]. In NSS-LC, Local Competition evaluates individuals against the current population and the novelty archive, rewarding those that outperform their behavioural neighbours [8]. One of NSS-LC's strengths is that surprise search can backtrack and revisit parts of the search space, which, combined with NSLC, balances both greedily searching for entirely new solutions and searching locally for highly fit solutions [8]. NSS-LC has achieved success, finding more solutions with fewer evaluations than other algorithms, when tested in 60 different maze environments [8].

3.3.3 Dominated Novelty Search. Dominated Novelty Search (DNS) is a hybrid evolutionary algorithm that uses a new local competition strategy which eliminates the need for "predefined bounds, grid structures, or fixed distance thresholds" by dynamically adapting to the distribution of the solutions and the structure and shape of the descriptor space [6]. Unlike previous methods which rely on distance thresholds or spatial partitioning, DNS creates a naturally developing competition structure based on proximity and relative solution performance, rewarding solutions that outperform their neighbours or discover novel behaviours relative to better-performing solutions [6]. DNS is highly effective for solving problems with intricate, multidimensional descriptor spaces [6]. DNS eliminates solutions if they are subpar in fitness or behaviourally similar to solutions that already exist, using a competition function that computes modified fitness values based on the distance between solutions and other better-performing solutions [6]. This competition mechanism results in two evolutionary pressures, requiring solutions to improve their fitness or discover unique behaviours that are different from better-performing solutions [6].

3.4 Task Environments

The experiment environments used in this project will all be extensions of the simulated environment used in [10].

In particular, the hybrid evolutionary methods experiment will use the Open Field Sheep-Herding environment from [10] as a benchmark and will also extend the maze environment to create three dynamic maze environments, as shown in Figure 1. The Easy maze environment will have the target zone (yellow circle) switching to the opposite corner mid-simulation. The Medium maze environment will have one of the sides of the maze becoming a cliff edge towards the end of the simulation, which sheep and dog agents can fall off. The Difficult maze environment will have dynamic blocks that randomly appear and disappear, intermittently blocking movement. Dog and sheep agents will move at a constant speed.

The body-brain experiment will be run in five environments. The control environment will be an open field with no changes in friction that would affect the motion of the robots. Complexity will be introduced by adding areas with different levels of surface friction, where areas with high friction (mud) will impede the robots' movement and areas with low friction (ice) will reduce traction. Environmental complexity is determined by the amount of ice and

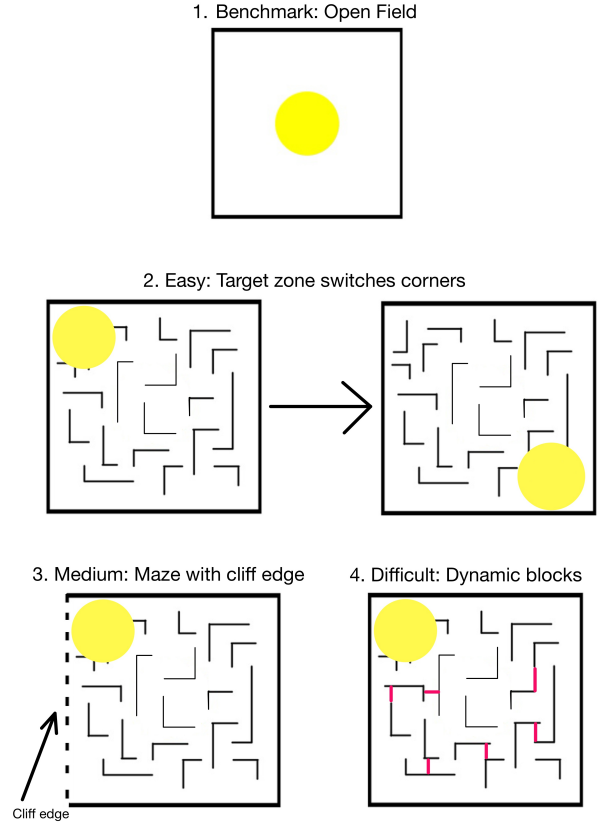


Figure 1: Task environments for hybrid evolutionary methods experiments.

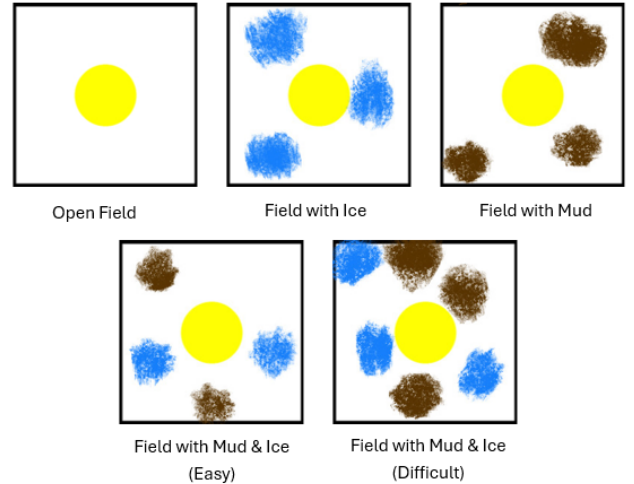


Figure 2: Task environments for body-brain-based experiments.

mud included in the terrain [2]. The environments are shown in Figure 2, where blue and brown patches represent areas covered in ice and mud, respectively, and the yellow circle represents the target zone.

3.5 Overview of Experiments

We will extend the code base from [10] for our experiments to answer our collective behaviour evolution research questions.

With the aim of answering research questions 3 and 4, each of the algorithms (SSGA, NSS-LC and DNS) will be tested across all four environments shown in Figure 1 (Section 3.4). Performance will be measured using number of sheep captured (exploitation) and proportion of maze explored (exploration), as mentioned in Table 1.

The body-brain-based experiments will be run in the environments depicted in Figure 2. For each environment, we will conduct two experiments: one with fixed dog morphologies and one with evolving dog morphologies. The performance of both groups will be compared and used to support the claim that agents with evolving morphology perform better than those with fixed morphology in complex tasks. Furthermore, the resulting agents' complexity will be compared and those findings will be used to answer research question 2. Section 3.6.2 describes the performance and complexity metrics to be used.

3.6 Evaluation Metrics

3.6.1 Hybrid Evolutionary Methods Experiments. For each of the difficulty levels, SSGA, NSS-LC and DNS will be run and two performance metrics will be recorded: the number of sheep captured and the proportion of the maze explored. This will be done for all four experiments, involving each of the three methods, SSGA, NSS-LC and DNS. Pairwise performance comparisons between each pair of algorithms (SSGA vs NSS-LC, SSGA vs DNS, NSS-LC vs DNS) will be conducted, using independent pairwise t-tests to determine whether differences in performance are significant across both metrics at each level of difficulty [14].

3.6.2 Body-Brain Based Experiments. Each environment will be used to run two experiments: one where the dog morphologies are fixed and one where the dog morphologies are evolving. In both experiments performance will be evaluated based on the ratio of captured sheep to total sheep [9], where a higher ratio indicates better performance. Mann-Whitney U-tests will be used to assess for notable differences in task performance between the two groups [15]. Additionally, morphological complexity will be compared and calculated as a weighted sum using the FOV and range values of the agents' sensors [7], and their maximum speed and body size.

4 ETHICAL, PROFESSIONAL AND LEGAL ISSUES

Our experiments do not require human involvement, and the anticipated results of these experiments do not create ethical issues. Roborobo! is open source and thus does not pose any legal issues. We will use the Roborobo! framework and inherited codebase in a professional manner.

5 RELATED WORK

This section provides more information on the sheep-herding environment we will be extending, as well as additional reasoning as to why DNS and NSS-LC were selected for this experiment over other hybrid evolutionary algorithms. It also details prior research on body-brain evolution in the context of swarm robotics.

5.1 Maze environment

As illustrated in Figures 1 and 2, we will extend the 2D bounded Sheep-Herding environment from [10], which features 20 dog agents tasked with herding 20 sheep agents into a target zone, simulated using the Roborobo! multi-agent simulator [5]. This environment serves as a model for various collective robotic tasks, such as search and rescue, making it useful for evaluating the effectiveness of evolutionary algorithms in evolving collective behaviour in complex environments [10].

5.2 Hybrid Evolutionary algorithms in adaptive multi-agent systems

Two foundational hybrid evolutionary algorithms which have achieved success are Novelty Search with Local Competition (NSLC) and MAP-Elites, however these algorithms have limitations [6]. NSLC has achieved success in that it was found to discover more "functional morphological diversity" in a single run than models that use global competition [13]. However, NSLC is limited in that it requires fixed distance parameters that need careful tuning, frequently needs complex adaptation mechanisms and is computationally expensive, due to computing the feature distance to all other agents [19] [6]. In addition, MAP-Elites, while being found to perform significantly better than optimization algorithms and other Quality Diversity algorithms, is limited by its need for pre-defined parameters and bounds for the feature space, which is especially limiting when the behaviour space is not known in advance [19] [6].

DNS and NSS-LC were selected for this experiment as they have certain advantages over NSLC and MAP-Elites. For instance, DNS eliminates the need for "predefined bounds, grid structures, or fixed distance thresholds" [6]. In addition, DNS was found to significantly outperform MAP-Elites in two of three domains where the descriptor space was well defined, but had implicit constraints, as well as significantly outperforming existing approaches across Quality Diversity benchmarks and maintaining an advantage in complex, multi-dimensional and unsupervised problems [6]. NSS-LC is advantaged over NSLC, as it has the ability to backtrack and revisit parts of the search space and when tested against other Quality Diversity algorithms over 60 different procedurally generated mazes, it was found to be "far more robust and efficient" [8].

5.3 Body-Brain Co-evolution in Swarm Robotics

Swarm robotics concerns multi-agent systems inspired by collective behaviours in nature, such as insect colonies, to develop decentralised robotic systems. Evolutionary swarm robotics applies evolutionary computation to optimise swarm behaviours and achieve diverse, adaptive collective behaviours suited to complex tasks [3]. These systems consist of numerous relatively simple robots

Experiment	Methods compared	Task environment used	Evaluation metric	Research objectives being addressed
1	Fixed Morphology vs Evolving Morphology	Open field	- Number of sheep captured - Morphological complexity	1 and 2
2	Fixed Morphology vs Evolving Morphology	Open field with ice	- Number of sheep captured - Morphological complexity	1 and 2
3	Fixed Morphology vs Evolving Morphology	Open field with mud	- Number of sheep captured - Morphological complexity	1 and 2
4	Fixed Morphology vs Evolving Morphology	Open field with mud and ice	- Number of sheep captured - Morphological complexity	1 and 2
5	Fixed Morphology vs Evolving Morphology	Open field with increased amount of mud and ice	- Number of sheep captured - Morphological complexity	1 and 2
6	DNS vs NSS-LC vs SSGA	Open field	Exploitation: number of sheep captured Exploration: proportion of maze explored	3 and 4
7	DNS vs NSS-LC vs SSGA	Maze with target zone that changes position	Exploitation: number of sheep captured Exploration: proportion of maze explored	3 and 4
8	DNS vs NSS-LC vs SSGA	Maze with cliff edge	Exploitation: number of sheep captured Exploration: proportion of maze explored	3 and 4
9	DNS vs NSS-LC vs SSGA	Maze with dynamic blocks	Exploitation: number of sheep captured Exploration: proportion of maze explored	3 and 4

Table 1: Experimental overview

whose interactions produce emergent group behaviours. Evolutionary swarm robotics faces unique challenges compared to single-robot systems, including coordinating decentralised behaviours and managing inter-agent interactions. A study by Mkhathshwa investigated morphology-controller co-evolution in swarm robotics. The researchers evaluated swarms of 100 robots tasked with cooperatively gathering resources in increasingly complex environments using a custom evolutionary embodied collective robotics simulator [17]. They tested three evolutionary strategies: behavioural diversity maintenance, morphology-behaviour diversity maintenance

and coupled behaviour-morphology diversity maintenance. The findings indicated that morphological diversity significantly enhanced swarm performance, particularly as complexity increased. The Double-Map EDQD-M (*Environment Driven Quality Diversity – coupled robot behaviour-morphology adaptation*) method allowed body and brain to evolve in parallel. This method produced the most efficient and adaptable swarm behaviours, demonstrating the importance of body-brain co-evolution in decentralised robotic systems [18].

6 ANTICIPATED OUTCOMES

This section outlines the expected results, research impact and contributions of the project.

6.1 Experiment Platform

The experiments rely on the successful extension of the inherited code base [10] and successful implementation of the relevant evolutionary algorithms. Thus, it is crucial to complete the environment extension in time to allow sufficient cluster runtime for the experiments.

6.2 Research and Impact

We anticipate that for the hybrid evolutionary methods experiment, the hybrid evolutionary algorithms will significantly outperform the objective-based algorithm, building on the findings from [9], where MAP-Elites outperformed SSGA, due to SSGA's lack of diversity in solution. In addition, [8] used 60 procedurally generated mazes to evaluate the effectiveness of hybrid evolutionary algorithms, with objective search as a baseline; however, objective search was unable to find a solution to any of the mazes. These prior results suggest that hybrid methods, such as DNS and NSS-LC, may be more effective than SSGA in complex, dynamic environments.

We hope to produce significant results regarding the use of hybrid evolutionary algorithms within a multi-agent system to solve a task in a dynamic environment. This would be beneficial to the field of hybrid evolutionary algorithms in MAS, as there is a lack of research on testing such algorithms in dynamic environments. If NSS-LC and/or DNS significantly outperform SSGA, as well as perform well overall in these dynamic environments, it will suggest that these algorithm(s) can be used in other complex and dynamic problems.

We expect that the body-brain based experiments will support previous research which suggests that for complex tasks collective robot systems with evolving morphology perform better and those with fixed morphology perform better in simple tasks [17]. Furthermore, we hope that our results show that increased complexity in the environment leads to more complex controllers and morphologies, as shown in previous studies [1, 2, 16].

6.3 Contributions

- Extension of the code base to include the three dynamic maze environments.
- Extension of the code to include SSGA, DNS and NSS-LC.
- Extension of the code base to include two new terrain textures.
- Extension of the code to allow for morphological representation and evolution.
- All code and research will be made publicly accessible to assist with future research.
- Results will extend previous work [9, 10] and contribute to the state of the art in exploration-exploitation evolutionary optimisation and body-brain co-evolution methods running within collective robotics applications.

7 PROJECT PLAN

This section presents the project's operational framework.

7.1 Risks

The risk matrix in Appendix A outlines the risks of the project.

7.2 Timeline

The project will run from the week of May 12th, with the final paper due on September 5th, and the project showcase in the week of October 27th. The scheduling, division of work, and important milestones are indicated in the Gantt chart in Appendix B.

7.3 Resources Required

- UCT High Performance Cluster ¹
- CHPC Cluster ²
- Roborobo! ³
- Access to a personal computer capable of running Linux, with a suitable IDE for development

7.4 Deliverables

The two final papers are the main deliverables for the project. Emily Raine's final paper will analyse the effectiveness of two hybrid evolutionary methods, NSS-LC and DNS, against a traditional objective method, SSGA in dynamic, complex environments. Naheed Khan's final paper will compare the performance of robots with evolving morphologies with those of fixed morphologies in a collective herding task, and evaluate the resulting morphologies and behaviours evolved in environments of varying complexities. The other deliverables and their deadlines are:

- Two Literature Reviews (28/03/2025)
- Project Proposal (6/05/2025)
- Project Progress Demonstration (28/07/2025 - 1/08/2025)
- Extension of code base (30/06/2025)
- Analysis of results (19/07/2025)
- Complete drafts of final papers (26/08/2025)
- Final Project Papers (5/09/2025)
- Project Code Final Submission (15/09/2025)
- Final Project Demonstration (22/09/2025 - 26/09/2025)
- Project Poster (6/10/2025)
- Project Website (13/10/2025)
- School of IT Showcase (27/10/2025)

7.5 Milestones

Notable milestones include the successful extension of the environment from [10], such that both experiments can be run, as well as submitting the drafts of the final papers and the final papers themselves. Another notable milestone is the final project demonstration. The Gantt chart in Appendix B outlines the project milestones.

7.6 Work Allocation

The work of extending the code base will be split equally between both team members, and together we will incorporate SSGA. Additionally, Emily Raine will extend the code base to include three

¹<https://ucthpc.uct.ac.za/>

²<https://www.chpc.ac.za/>

³<https://github.com/nekonaute/roborobo4>

dynamic maze environments and the two hybrid algorithms. Naheed Khan will extend the code base by introducing morphological evolution in dog agents and adding five new environments of varying complexity. Furthermore, Naheed Khan will perform the experiment on body-brain evolution in simulated collective herding environments to answer research questions 1 and 2. Emily Raine will perform the experiment on hybrid evolutionary methods in dynamic collective herding systems to answer research questions 3 and 4. We will each write our own final report.

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A RISK MATRIX

Risk	Probability	Impact	Mitigation
Updating the code environment is timely, taking longer than expected, which delays the rest of the project.	Medium	High	Both team members will follow the Gantt chart closely and we will budget time upfront for issues and delays. We will also have weekly meetings to ensure that we remain on track.
Team member fails to meet deadline.	Low	High	We will do additional planning within the time period allocated for each stage, to ensure that we don't fall behind.
Failure to extend the environment.	Low	High	Team members will familiarise themselves with the Roborobo! Framework and existing codebase as soon as possible to identify potential pitfalls and research relevant methods that will be used.
Unsuccessful implementation of new algorithms (DNS and NSS-LC) to evolve dog agents.	Low	High	Obtain a clear understanding of the inherited code and algorithms. Perform thorough research on how to implement the algorithms as part of the existing code base.
Unsuccessful evolution of dog morphology.	Medium	High	Obtain sufficient knowledge on how to maximise chances of successful evolution and research different approaches.
Failure to address research objectives due to lack of experience with neural networks/evolutionary algorithms.	Medium	Medium	Research common obstacles when working with evolutionary algorithms and neural networks to ensure team members are adequately prepared to handle problems and troubleshoot.
Cluster is unavailable when needed to run experiments, resulting in the team falling behind schedule.	Medium	High	Ensure extensions to the environment and algorithmic frameworks are implemented with sufficient time left to account for scheduling delays.

B GANTT CHART

